

Inhibition of bacterial quorum sensing by vanilla extract.

AIMS: The purpose of this study was to search for a novel quorum sensing inhibitor and analyse its inhibitory activity. **METHODS AND RESULTS:** Quorum sensing inhibition was monitored using the Tn-5 mutant, *Chromobacterium violaceum* CV026. Vanilla beans (*Vanilla planifolia* Andrews) were extracted using 75% (v/v) aqueous methanol and added to *C. violaceum* CV026 cultures. Inhibitory activity was measured by quantifying violacein production using a spectrophotometer. The results have revealed that vanilla extract significantly reduced violacein production in a concentration-dependent manner, indicating inhibition of quorum sensing. **CONCLUSIONS:** Vanilla, a widely used spice and flavour, can inhibit bacterial quorum sensing. **SIGNIFICANCE AND IMPACT OF THE STUDY:** The results suggest that the intake of vanilla-containing food materials might promote human health by inhibiting quorum sensing and preventing bacterial pathogenesis. Further studies are required to isolate specific substances from vanilla extract acting as quorum sensing inhibitors.